

HE2001 Microeconomics II

Midterm Practice 1 (Solutions)

Academic Year 2025/2026, Semester 2

Quantitative Research Society @NTU

August 17, 2026

Overview

This mock exam focuses on core tools in microeconomic theory and asset-pricing foundations:

- revealed preference (direct revealed preference, WARP);
- Marshallian/Hicksian demand and Slutsky decomposition;
- labor–leisure choice and compensated labor supply;
- present value and net present value calculations;
- stochastic dominance and expected utility under risk;
- state prices, contingent claims, and arbitrage pricing;
- intertemporal choice with borrowing/saving.

Instructions

- This exam consists of 6 short questions (8 marks each) and 3 long questions (20 marks each).
- Total marks: 108.
- Show all working clearly. Answers without justification may receive zero marks.
- The notation (x_1, x_2) denotes a bundle of goods 1 and 2.
- The notation (p_1, p_2) denotes prices of goods 1 and 2.

Part I

Short Questions

Question 1 (8 marks)

Alice maximizes a strictly increasing utility function. You observe the following purchasing behavior:

- At prices $p^1 = (2, 1)$, Alice buys $x^1 = (3, 6)$
- At prices $p^2 = (1, 2)$, Alice buys $x^2 = (5, 4)$
- At prices $p^3 = (3, 2)$, Alice buys $x^3 = (2, 5)$

- (2 marks) Calculate how much money Alice spent in each period.
- (3 marks) Determine all direct revealed preference relations. That is, for each pair of bundles, determine if one is directly revealed preferred to the other.
- (3 marks) Does Alice's behavior satisfy WARP? If not, identify the violation. Does her behavior satisfy GARP? Justify your answer.

Solution

Method 1: Direct affordability calculations

- Expenditures.** Let $m^t := p^t \cdot x^t$.

$$m^1 = (2, 1) \cdot (3, 6) = 12,$$

$$m^2 = (1, 2) \cdot (5, 4) = 13,$$

$$m^3 = (3, 2) \cdot (2, 5) = 16.$$

Hence

$$\boxed{m^1 = 12, \quad m^2 = 13, \quad m^3 = 16.}$$

- Direct revealed preference relations.**

Recall:

$$x^t R x^s \iff p^t \cdot x^s \leq p^t \cdot x^t, \quad x^t P x^s \iff p^t \cdot x^s < p^t \cdot x^t.$$

Cross-costs:

$$\begin{aligned}
 p^1 \cdot x^2 = 14 > 12 &\Rightarrow \neg(x^1 R x^2), \\
 p^1 \cdot x^3 = 9 < 12 &\Rightarrow x^1 P x^3; \\
 p^2 \cdot x^1 = 15 > 13 &\Rightarrow \neg(x^2 R x^1), \\
 p^2 \cdot x^3 = 12 < 13 &\Rightarrow x^2 P x^3; \\
 p^3 \cdot x^1 = 21 > 16 &\Rightarrow \neg(x^3 R x^1), \\
 p^3 \cdot x^2 = 23 > 16 &\Rightarrow \neg(x^3 R x^2).
 \end{aligned}$$

Thus the only nontrivial direct revealed preference relations are

$$\boxed{x^1 R x^3, x^2 R x^3}$$

and in fact strictly,

$$\boxed{x^1 P x^3, x^2 P x^3}.$$

(c) **WARP and GARP.**

WARP violation requires $\exists t \neq s$ such that

$$x^t R x^s \text{ and } x^s P x^t.$$

No such pair exists, so

$$\boxed{\text{WARP holds.}}$$

Let R^* be the transitive closure of R . The only nontrivial R^* relations are

$$x^1 R^* x^3, \quad x^2 R^* x^3.$$

GARP violation requires $\exists t, s$ such that

$$x^t R^* x^s \text{ and } x^s P x^t.$$

Check the only candidates ($s = 3$):

$$p^3 \cdot x^1 = 21 > 16 = p^3 \cdot x^3, \quad p^3 \cdot x^2 = 23 > 16 = p^3 \cdot x^3.$$

Hence no GARP violation:

$$\boxed{\text{GARP holds.}}$$

Method 2: Concise R, P, R^* test

(a)

$$m^1 = p^1 x^1 = 12, \quad m^2 = p^2 x^2 = 13, \quad m^3 = p^3 x^3 = 16.$$

(b) Define

$$x^t R x^s \iff p^t x^s \leq p^t x^t, \quad x^t P x^s \iff p^t x^s < p^t x^t.$$

Compute matrix $(p^t \cdot x^s)_{t,s}$:

$$\begin{pmatrix} 12 & 14 & 9 \\ 15 & 13 & 12 \\ 21 & 23 & 16 \end{pmatrix}.$$

Compare each row to its diagonal entry (12, 13, 16):

$$R = \{(1, 1), (2, 2), (3, 3), (1, 3), (2, 3)\},$$

$$P = \{(1, 3), (2, 3)\}.$$

Hence (nontrivial):

$$\boxed{x^1 R x^3, x^2 R x^3}, \quad \boxed{x^1 P x^3, x^2 P x^3}.$$

(c) **WARP:**

$$\text{WARP violated} \iff \exists t \neq s : (t, s) \in R, (s, t) \in P.$$

But $P = \{(1, 3), (2, 3)\}$ and neither $(3, 1)$ nor $(3, 2)$ lies in R . Therefore

$$\boxed{\text{WARP holds.}}$$

GARP: Let R^* be the transitive closure of R . Since there is no chain creating new off-diagonal links,

$$R^* = R.$$

A GARP violation is

$$\exists t, s : (t, s) \in R^* \text{ and } p^s x^t < p^s x^s.$$

Only off-diagonal candidates are $(1, 3), (2, 3)$. Check:

$$p^3 x^1 = 21 > 16 = p^3 x^3, \quad p^3 x^2 = 23 > 16 = p^3 x^3.$$

So no violation:

$$\boxed{\text{GARP holds.}}$$

Marking scheme (8 marks)**(a) (2 marks)** Expenditures

- 1 mark: correct computation of at least two expenditures / correct method $m^t = p^t \cdot x^t$.
- 1 mark: all three correct values $m^1 = 12$, $m^2 = 13$, $m^3 = 16$.

(b) (3 marks) Direct revealed preference relations

- 1 mark: correct criterion $x^t R x^s \iff p^t \cdot x^s \leq p^t \cdot x^t$.
- 1 mark: correct cross-cost comparisons (or equivalent table/matrix).
- 1 mark: correct conclusion that only $x^1 R x^3$ and $x^2 R x^3$ hold (strictly).

(c) (3 marks) WARP and GARP

- 1.5 marks: correct WARP conclusion with valid justification (no two-way contradiction).
- 1.5 marks: correct GARP conclusion with valid justification (no violating chain / affordability contradiction).

Question 2: Leontief demand and Slutsky decomposition

Consider a consumer with utility function $U(x_1, x_2) = \min\{3x_1, x_2\}$ who faces prices $(p_1, p_2) = (2, 3)$ and has income $m = 30$.

- (3 marks) Find the consumer's optimal consumption bundle. Show your working including the condition for optimality.
- (2 marks) The price of good 1 increases to $p'_1 = 3$. Find the new optimal consumption bundle.
- (3 marks) Calculate the income and substitution effects for good 1 using the Slutsky method. Are the goods normal or inferior?

Solution

- (a) **Initial optimal bundle.**

Key idea. For $U(x_1, x_2) = \min\{3x_1, x_2\}$, utility increases only when both components rise in the fixed ratio $x_2 = 3x_1$.

Under $(p_1, p_2) = (2, 3)$ and $m = 30$, the budget set is $2x_1 + 3x_2 \leq 30$. Optimality for Leontief preferences requires the kink condition:

$$3x_1 = x_2$$

Substitute $x_2 = 3x_1$ into the binding budget constraint $2x_1 + 3x_2 = 30$:

$$2x_1 + 3(3x_1) = 2x_1 + 9x_1 = 11x_1 = 30 \implies x_1^* = \frac{30}{11}, \quad x_2^* = 3x_1^* = \frac{90}{11}$$

$$(x_1^*, x_2^*) = \left(\frac{30}{11}, \frac{90}{11} \right)$$

- (b) **New optimal bundle.**

New prices are $(p'_1, p_2) = (3, 3)$, income $m = 30$, so the new budget constraint is $3x_1 + 3x_2 \leq 30$. The optimality condition remains $x_2 = 3x_1$ at the optimum. Substituting this into the new budget constraint:

$$3x_1 + 3(3x_1) = 12x_1 = 30 \implies x'_1 = \frac{5}{2}, \quad x'_2 = 3x'_1 = \frac{15}{2}$$

$$(x'_1, x'_2) = \left(\frac{5}{2}, \frac{15}{2} \right)$$

(c) **Income and substitution effects.****Method 1: Slutsky (Compensated) Decomposition**

Let the original optimal bundle be $x = (x_1, x_2) = \left(\frac{30}{11}, \frac{90}{11}\right)$. The Slutsky compensation (income that makes x affordable at the new prices) is:

$$m^s = p'_1 x_1 + p_2 x_2 = 3 \left(\frac{30}{11}\right) + 3 \left(\frac{90}{11}\right) = \frac{360}{11}$$

At prices $(3, 3)$ and income m^s , the Leontief kink $x_2 = 3x_1$ and budget $3x_1 + 3x_2 = m^s$ imply:

$$12x_1^s = \frac{360}{11} \implies x_1^s = \frac{30}{11}$$

Hence, the Slutsky substitution and income effects are:

$$\Delta x_1^s = x_1^s - x_1^* = \frac{30}{11} - \frac{30}{11} = 0$$

$$\Delta x_1^n = x'_1 - x_1^s = \frac{5}{2} - \frac{30}{11} = -\frac{5}{22}$$

$\Delta x_1^s = 0, \quad \Delta x_1^n = -\frac{5}{22}, \quad \text{Good 1 is a Normal Good}$
--

Method 2: Hicksian Demand (Expenditure Minimisation)

Key idea. For Leontief utility, Hicksian (compensated) demand is derived via expenditure minimisation. Because the optimal consumption bundle is strictly pinned to the vertex of the L-shaped indifference curves, relative price changes do not alter the compensated quantity ratio, resulting in a zero substitution effect.

Step 1: Formulate the Expenditure Minimisation Problem (EMP). To achieve a target utility level $\bar{u}$, the consumer minimises total expenditure:

$$\min_{x_1, x_2} p_1 x_1 + p_2 x_2 \quad \text{subject to} \quad \min\{3x_1, x_2\} \geq \bar{u}$$

For Leontief preferences, any expenditure off the vertex is suboptimal. Therefore, the constraint must bind exactly at the kink:

$$3x_1 = x_2 = \bar{u}$$

Step 2: Derive the Hicksian demands. Solving the vertex condition yields Hicksian demand functions that are independent of prices:

$$x_1^h(p_1, p_2, \bar{u}) = \frac{\bar{u}}{3}, \quad x_2^h(p_1, p_2, \bar{u}) = \bar{u}$$

Step 3: Calculate the Substitution Effect (SE). Find the initial utility level at the original optimal bundle $(x_1^*, x_2^*) = (\frac{30}{11}, \frac{90}{11})$:

$$\bar{u} = U\left(\frac{30}{11}, \frac{90}{11}\right) = \min\left\{3\left(\frac{30}{11}\right), \frac{90}{11}\right\} = \frac{90}{11}$$

The compensated demand for good 1 at the new prices p' and target utility $\bar{u}$ is:

$$x_1^h(p'_1, p_2, \bar{u}) = \frac{90/11}{3} = \frac{30}{11}$$

The substitution effect is the change in Hicksian demand:

$$\Delta x_1^s = x_1^h(p'_1, p_2, \bar{u}) - x_1^* = \frac{30}{11} - \frac{30}{11} = 0$$

Step 4: Calculate the Income Effect (IE). The income effect is the change from the compensated demand to the new uncompensated (Marshallian) demand $x'_1 = \frac{5}{2}$:

$$\Delta x_1^n = x'_1 - x_1^h(p'_1, p_2, \bar{u}) = \frac{5}{2} - \frac{30}{11} = \frac{55 - 60}{22} = -\frac{5}{22}$$

Step 5: Determine if the good is normal or inferior. The price of good 1 increased, which reduces the consumer's real income. Since the income effect is negative ($\Delta x_1^n < 0$), the decrease in real income led to a decrease in the quantity demanded. Because consumption moves in the same direction as real income, good 1 is a **normal good**.

$\Delta x_1^s = 0, \quad \Delta x_1^n = -\frac{5}{22}, \quad \text{Good 1 is a Normal Good}$
--

Marking scheme (8 marks)**(a) (3 marks)** Initial optimal bundle

- 1 mark: Identifying the Leontief kink condition $x_2 = 3x_1$.
- 1 mark: Substituting the condition into the binding budget constraint $2x_1 + 3x_2 = 30$.
- 1 mark: Correct optimal bundle $(x_1^*, x_2^*) = (\frac{30}{11}, \frac{90}{11})$.

(b) (2 marks) New optimal bundle

- 1 mark: Setting up the new budget constraint $3x_1 + 3x_2 = 30$ and applying the kink condition.
- 1 mark: Correct new optimal bundle $(x_1', x_2') = (\frac{5}{2}, \frac{15}{2})$.

(c) (3 marks) Income and substitution effects

- 1 mark: Correct setup for finding the compensated demand (calculating Slutsky income $m^s = \frac{360}{11}$ OR Hicksian target utility $\bar{u} = \frac{90}{11}$).
- 1 mark: Correct computation of the substitution effect ($\Delta x_1^s = 0$) and income effect ($\Delta x_1^i = -\frac{5}{22}$).
- 1 mark: Correct conclusion that good 1 is a normal good, with valid justification (demand moves in the same direction as real income).

Question 3: Labor supply, wage changes, and Slutsky effects

Consider a worker who maximizes utility from consumption (c) and leisure (ℓ) given by $U(c, \ell) = c^{1/2}\ell^{1/2}$. The worker has 16 hours available per day. The wage rate is w dollars per hour and she has no non-labor income.

- (3 marks) Write down the budget constraint and find the optimal labor supply function $L^*(w)$ where L is hours worked.
- (2 marks) At wage $w = \$10$, how many hours does the worker supply? What is her consumption?
- (3 marks) The wage increases to $w' = \$20$. Calculate the income and substitution effects on labor supply. Does labor supply increase or decrease? Explain why.

Solution

Method 1: Direct optimization in L (Marshallian + Hicksian decomposition)

Let total time be $T = 16$. If L is hours worked, then

$$\ell = 16 - L, \quad c = wL, \quad 0 \leq L \leq 16.$$

- Budget constraint and optimal labor supply.**

Budget (equivalently):

$$c = wL, \quad \ell = 16 - L$$

or in (c, ℓ) -form,

$$\boxed{c + w\ell = 16w.}$$

Utility as a function of L :

$$U(L) = \sqrt{wL}\sqrt{16 - L}.$$

Since $w > 0$, maximize equivalently

$$U(L)^2 = wL(16 - L) \iff \max_{L \in [0, 16]} f(L) := L(16 - L).$$

FOC/SOC:

$$f'(L) = 16 - 2L = 0 \Rightarrow L = 8, \quad f''(L) = -2 < 0.$$

Hence

$$\boxed{L^*(w) = 8 \quad (\forall w > 0).}$$

(b) **At** $w = 10$.

$$L^*(10) = 8, \quad c = wL = 10 \cdot 8 = 80.$$

Thus

$$\boxed{L = 8, \quad c = 80.}$$

(c) **Wage rises to $w' = 20$: substitution and income effects on labor supply.**

Marshallian labor supply:

$$L^*(20) = 8 \quad \Rightarrow \quad \Delta L = 8 - 8 = 0.$$

To decompose, hold initial utility fixed. At $w = 10$, the chosen bundle is

$$(c_0, \ell_0) = (80, 8), \quad u_0 = U(c_0, \ell_0) = \sqrt{80 \cdot 8} = 8\sqrt{10}.$$

Hicksian problem at wage w :

$$\min_{c, \ell \geq 0} c + w\ell \quad \text{s.t.} \quad \sqrt{c\ell} \geq u.$$

At optimum (Cobb–Douglas tangency + binding constraint):

$$c = w\ell, \quad c\ell = u^2.$$

Hence

$$w\ell^2 = u^2 \Rightarrow \ell^h(w, u) = \frac{u}{\sqrt{w}}, \quad L^h(w, u) = 16 - \frac{u}{\sqrt{w}}.$$

So at $w' = 20$, holding u_0 fixed:

$$L^h(20, u_0) = 16 - \frac{8\sqrt{10}}{\sqrt{20}} = 16 - \frac{8\sqrt{10}}{2\sqrt{5}} = 16 - 4\sqrt{2}.$$

Therefore (labor supply version):

$$\Delta L^s = L^h(20, u_0) - L^*(10) = (16 - 4\sqrt{2}) - 8 = \boxed{8 - 4\sqrt{2}},$$

$$\Delta L^n = L^*(20) - L^h(20, u_0) = 8 - (16 - 4\sqrt{2}) = \boxed{-8 + 4\sqrt{2}}.$$

Thus

$$\Delta L = \Delta L^s + \Delta L^n = 0.$$

Interpretation: the substitution effect raises labor supply ($\Delta L^s > 0$), while the income effect lowers labor supply by exactly the same amount ($\Delta L^n < 0$); hence labor supply is unchanged.

Labor supply does not change (stays at 8 hours).

Method 2: Full-income Cobb–Douglas

(a) Since $c = wL$ and $\ell = 16 - L$,

$$c + w\ell = wL + w(16 - L) = 16w.$$

So

$$\max_{c, \ell \geq 0} c^{1/2} \ell^{1/2} \quad \text{s.t.} \quad c + w\ell = 16w.$$

For Cobb–Douglas with exponents $(1/2, 1/2)$, budget shares are $1/2, 1/2$:

$$c^*(w) = \frac{1}{2}(16w) = 8w, \quad w\ell^*(w) = \frac{1}{2}(16w) = 8w \Rightarrow \ell^*(w) = 8.$$

Hence

$$\boxed{L^*(w) = 16 - \ell^*(w) = 8.}$$

(b) At $w = 10$:

$$L^* = 8, \quad c^* = 8w = 80.$$

So

$$\boxed{L = 8, \quad c = 80.}$$

(c) At $w' = 20$, Marshallian labor supply remains

$$L^*(20) = 8 \Rightarrow \Delta L = 0.$$

For Hicksian demands of $u(c, \ell) = \sqrt{c\ell}$:

$$\ell^h(w, u) = \frac{u}{\sqrt{w}} \Rightarrow L^h(w, u) = 16 - \frac{u}{\sqrt{w}}.$$

Initial utility:

$$u_0 = \sqrt{80 \cdot 8} = 8\sqrt{10}.$$

Thus

$$L^h(20, u_0) = 16 - \frac{8\sqrt{10}}{\sqrt{20}} = 16 - 4\sqrt{2}.$$

So

$$\begin{aligned} \Delta L^s &= L^h(20, u_0) - L^*(10) = 8 - 4\sqrt{2}, \\ \Delta L^n &= L^*(20) - L^h(20, u_0) = -8 + 4\sqrt{2}, \end{aligned}$$

$$\boxed{\Delta L^s + \Delta L^n = 0, \quad \Delta L = 0.}$$

Hence labor supply *does not* increase/decrease overall.

Method 3: Leisure-demand first, then convert to labor(a) Solve in (c, ℓ) with

$$c + w\ell = 16w.$$

Lagrangian:

$$\mathcal{L}(c, \ell, \lambda) = c^{1/2}\ell^{1/2} + \lambda(16w - c - w\ell).$$

FOCs:

$$\frac{1}{2}c^{-1/2}\ell^{1/2} = \lambda, \quad \frac{1}{2}c^{1/2}\ell^{-1/2} = \lambda w.$$

Divide the first by the second:

$$\frac{\ell}{c} = \frac{1}{w} \Rightarrow c = w\ell.$$

Substitute into budget:

$$w\ell + w\ell = 16w \Rightarrow \ell^* = 8.$$

Therefore

$$\boxed{L^*(w) = 16 - \ell^* = 8.}$$

(b) At $w = 10$:

$$L = 8, \quad c = 10 \cdot 8 = 80.$$

So

$$\boxed{L = 8, \quad c = 80.}$$

(c) Wage increase $10 \rightarrow 20$:

$$L^*(10) = L^*(20) = 8 \Rightarrow \Delta L = 0.$$

Using Hicksian labor supply from Method 1/2,

$$L^h(w, u) = 16 - \frac{u}{\sqrt{w}}, \quad u_0 = 8\sqrt{10},$$

gives

$$L^h(20, u_0) = 16 - 4\sqrt{2}.$$

Hence

$$\boxed{\Delta L^s = 8 - 4\sqrt{2} > 0, \quad \Delta L^n = -8 + 4\sqrt{2} < 0, \quad \Delta L = 0.}$$

Marking scheme (suggested, 8 marks)

(a) **(3 marks)** Budget constraint and $L^*(w)$

- 1 mark: correct budget/time relations ($c = wL$, $\ell = 16 - L$, or $c + w\ell = 16w$).
- 1 mark: correct optimization setup and derivation.
- 1 mark: correct conclusion $L^*(w) = 8$.

(b) **(2 marks)** Numerical values at $w = 10$

- 1 mark: $L = 8$.
- 1 mark: $c = 80$.

(c) **(3 marks)** Slutsky decomposition for labor supply ($10 \rightarrow 20$)

- 1 mark: correct total effect $\Delta L = 0$ and/or correct compensated setup.
- 1 mark: correct substitution effect $\Delta L^s = 8 - 4\sqrt{2}$.
- 1 mark: correct income effect $\Delta L^n = -8 + 4\sqrt{2}$ and conclusion (offset exactly).

Question 4 (8 marks)

Consider an economy with three time periods ($t = 0, 1, 2$) and interest rate $r = 0.20$.

- (a) (2 marks) What is the present value of receiving \$100 in period 1 and \$144 in period 2?
- (b) (3 marks) Asset A pays (10, 20, 30) in periods 0, 1, and 2. Asset B pays (5, 40, 20). Which asset has higher present value? Show your calculation.
- (c) (3 marks) You can buy asset C for \$80 today. It pays \$0 in period 1 and \$120 in period 2. Should you buy asset C or save your money at the interest rate $r = 0.20$? Calculate the net present value of buying asset C .

Solution

Method 1: Discounted present value and net present value

Key idea. Discount each future cash flow by $(1 + r)^t$ and compare values at $t = 0$.

Here $r = 0.20$, so $(1 + r) = 1.2$ and $(1 + r)^2 = 1.44$.

- (a) Present value:

$$PV = \frac{100}{1.2} + \frac{144}{1.44} = \frac{250}{3} + 100 = \frac{550}{3}$$

$$PV = \frac{550}{3} \approx 183.33$$

- (b) Asset present values:

$$PV_A = 10 + \frac{20}{1.2} + \frac{30}{1.44} = 10 + \frac{50}{3} + \frac{125}{6} = \frac{285}{6} = 47.5$$

$$PV_B = 5 + \frac{40}{1.2} + \frac{20}{1.44} = 5 + \frac{100}{3} + \frac{125}{9} = \frac{470}{9} \approx 52.22$$

Thus $PV_B > PV_A$, so asset B has a higher present value.

$$PV_A = 47.5, \quad PV_B = \frac{470}{9} \approx 52.22, \quad \text{Asset } B \text{ is higher}$$

- (c) Asset C costs 80 today and pays 120 in period 2. Its present value is $\frac{120}{1.44} = \frac{250}{3} \approx 83.33$, so the net present value is:

$$NPV = \frac{120}{1.44} - 80 = \frac{250}{3} - 80 = \frac{10}{3} > 0$$

$$\text{Buy Asset } C \quad \left(NPV = \frac{10}{3} > 0 \right)$$

Method 2: Use discount factors and compare replicated value

Key idea. Let discount factors be $d_1 = \frac{1}{1.2} = \frac{5}{6}$ and $d_2 = \frac{1}{1.44} = \frac{25}{36}$; then $PV = \sum_t d_t \cdot \text{cashflow}_t$.

(a) Present value:

$$PV = 100d_1 + 144d_2 = 100 \left(\frac{5}{6} \right) + 144 \left(\frac{25}{36} \right) = \frac{250}{3} + 100 = \frac{550}{3}$$

$$PV = \frac{550}{3} \approx 183.33$$

(b) Asset present values:

$$PV_A = 10 + 20d_1 + 30d_2 = 10 + 20 \left(\frac{5}{6} \right) + 30 \left(\frac{25}{36} \right) = 10 + \frac{50}{3} + \frac{125}{6} = 47.5$$

$$PV_B = 5 + 40d_1 + 20d_2 = 5 + 40 \left(\frac{5}{6} \right) + 20 \left(\frac{25}{36} \right) = 5 + \frac{100}{3} + \frac{125}{9} = \frac{470}{9} \approx 52.22$$

$$PV_A = 47.5, \quad PV_B = \frac{470}{9} \approx 52.22, \quad \text{Asset } B \text{ is higher}$$

(c) Compare cost 80 with the present value of cash flows:

$$PV_C = 120d_2 = 120 \left(\frac{25}{36} \right) = \frac{250}{3} \approx 83.33$$

Since $80 < \frac{250}{3}$, buying C dominates saving at rate r .

$$NPV = \frac{250}{3} - 80 = \frac{10}{3} > 0$$

$$\text{Buy Asset } C \quad \left(NPV = \frac{10}{3} > 0 \right)$$

Marking scheme (8 marks)**(a) (2 marks)** Present value calculation

- 1 mark: Correct setup of the present value calculation (discount formula or discount factors).
- 1 mark: Correct final present value of $\frac{550}{3}$ (or approx 183.33).

(b) (3 marks) Asset present values comparison

- 1 mark: Correct calculation of $PV_A = 47.5$.
- 1 mark: Correct calculation of $PV_B = \frac{470}{9}$ (or approx 52.22).
- 1 mark: Correct conclusion that Asset B has a higher present value.

(c) (3 marks) Net present value and investment decision

- 1 mark: Correct calculation of the present value of Asset C 's payoff ($PV = \frac{250}{3}$ or approx 83.33).
- 1 mark: Correct calculation of the Net Present Value ($NPV = \frac{10}{3}$ or approx 3.33).
- 1 mark: Correct conclusion to buy Asset C , properly justified by a positive NPV or the PV being strictly greater than the cost.

Question 5: Stochastic dominance and expected utility

There are three equally likely states of nature. Consider three contingent consumption bundles:

- $x = (9, 6, 3)$
- $y = (6, 6, 6)$
- $z = (5, 7, 6)$

- (a) (2 marks) Calculate the expected value of each bundle.
- (b) (3 marks) Does bundle x first-order stochastically dominate bundle z ? Does bundle z first-order stochastically dominate bundle x ? Justify your answers.
- (c) (3 marks) A risk-averse consumer with utility $u(c) = \sqrt{c}$ must choose between bundles y and z . Which bundle will she choose? Show your calculation.

Solution

Method 1: Discrete distribution table + CDF checks + expected utility

Let each state have probability $1/3$. Treat each bundle as a discrete random variable.

- (a) **Expected values.**

$$\mathbb{E}[x] = \frac{9 + 6 + 3}{3} = 6, \quad \mathbb{E}[y] = \frac{6 + 6 + 6}{3} = 6, \quad \mathbb{E}[z] = \frac{5 + 7 + 6}{3} = 6.$$

Hence

$$\boxed{\mathbb{E}[x] = \mathbb{E}[y] = \mathbb{E}[z] = 6.}$$

- (b) **FOSD using discrete CDF table (check at 3, 5, 6, 7, 9).**

Write the discrete distributions:

$$X \in \{3, 6, 9\} \text{ with prob. } 1/3 \text{ each,} \quad Z \in \{5, 6, 7\} \text{ with prob. } 1/3 \text{ each.}$$

Thus the CDFs $F_x(t) = \Pr(X \leq t)$, $F_z(t) = \Pr(Z \leq t)$ at the relevant support points are:

t	3	5	6	7	9
$F_x(t)$	$\frac{1}{3}$	$\frac{1}{3}$	$\frac{2}{3}$	$\frac{2}{3}$	1
$F_z(t)$	0	$\frac{1}{3}$	$\frac{2}{3}$	1	1

Recall:

$$x \text{ FOSD } z \iff F_x(t) \leq F_z(t) \forall t, \quad z \text{ FOSD } x \iff F_z(t) \leq F_x(t) \forall t.$$

Now:

- At $t = 3$: $F_x(3) = \frac{1}{3} > 0 = F_z(3)$, so x *does not* FOSD z .
- At $t = 7$: $F_z(7) = 1 > \frac{2}{3} = F_x(7)$, so z *does not* FOSD x .

Therefore

Neither x nor z first-order stochastically dominates the other.

(c) **Choice between y and z for $u(c) = \sqrt{c}$.**

For $y = (6, 6, 6)$,

$$\mathbb{E}[u(y)] = \frac{1}{3}(\sqrt{6} + \sqrt{6} + \sqrt{6}) = \sqrt{6}.$$

For $z = (5, 7, 6)$,

$$\mathbb{E}[u(z)] = \frac{1}{3}(\sqrt{5} + \sqrt{7} + \sqrt{6}).$$

Numerically,

$$\begin{aligned}\mathbb{E}[u(y)] &= \sqrt{6} \approx 2.44949, \\ \mathbb{E}[u(z)] &= \frac{\sqrt{5} + \sqrt{7} + \sqrt{6}}{3} \approx 2.44377.\end{aligned}$$

Hence

$$\mathbb{E}[u(y)] > \mathbb{E}[u(z)].$$

So the consumer chooses

$$\boxed{y.}$$

Method 2: Concise stochastic-dominance test + Jensen for part (c)

(a)

$$\mathbb{E}[x] = \mathbb{E}[y] = \mathbb{E}[z] = 6.$$

(b) Let F_x, F_z be the CDFs. Using the discrete checks:

$$F_x(3) = \frac{1}{3} > 0 = F_z(3) \Rightarrow x \not\prec_{\text{FOSD}} z,$$

$$F_z(7) = 1 > \frac{2}{3} = F_x(7) \Rightarrow z \not\prec_{\text{FOSD}} x.$$

Thus

No FOSD relation between x and z .

(c) Since $y \equiv 6$ (certainty) and $\mathbb{E}[z] = 6$, and $u(c) = \sqrt{c}$ is strictly concave,

$$\mathbb{E}[u(z)] < u(\mathbb{E}[z]) = u(6) = \sqrt{6} = \mathbb{E}[u(y)].$$

Hence

$$\boxed{y \succ z.}$$

Method 3: Direct expected-utility comparison (fully computational)

(a)

$$\boxed{\mathbb{E}[x] = \mathbb{E}[y] = \mathbb{E}[z] = 6.}$$

(b) CDF crossing (using support points 3, 5, 6, 7, 9):

$$F_x(3) = \frac{1}{3} > 0 = F_z(3), \quad F_z(7) = 1 > \frac{2}{3} = F_x(7).$$

So neither FOSD holds.

(c)

$$EU(y) = \sqrt{6}, \quad EU(z) = \frac{\sqrt{5} + \sqrt{7} + \sqrt{6}}{3}.$$

Difference:

$$EU(y) - EU(z) = \frac{2\sqrt{6} - \sqrt{5} - \sqrt{7}}{3} > 0.$$

(Numerically ≈ 0.00572 .) Therefore
$$\boxed{\text{Choose } y.}$$
Marking scheme (suggested, 8 marks)(a) **(2 marks)** Expected values

- 1 mark: correct method (average over 3 equally likely states)
- 1 mark: correct values $\mathbb{E}[x] = \mathbb{E}[y] = \mathbb{E}[z] = 6$

(b) **(3 marks)** FOSD comparison of x and z

- 1 mark: correct use of CDF definition / criterion
- 1 mark: correct discrete CDF comparisons (e.g. at $t = 3$ and $t = 7$)
- 1 mark: correct conclusion that neither FOSD the other

(c) **(3 marks)** Expected utility under $u(c) = \sqrt{c}$

- 1 mark: correct $EU(y) = \sqrt{6}$
- 1 mark: correct $EU(z) = \frac{\sqrt{5} + \sqrt{7} + \sqrt{6}}{3}$
- 1 mark: correct comparison and conclusion (choose y)

Question 6: State prices and arbitrage

Consider two states (Good and Bad) with probabilities $\pi_G = 0.3$ and $\pi_B = 0.7$. Two assets are available:

- Asset 1: pays \$2 in the Good state and \$1 in the Bad state, costs \$1.30
- Asset 2: pays \$1 in the Good state and \$2 in the Bad state, costs \$1.60

- (3 marks) Find the state price vector (q_G, q_B) implied by these asset prices.
- (2 marks) What is the price of contingent consumption bundle $(5, 4)$?
- (3 marks) A new asset 3 is introduced that pays \$3 in both states. Using the state prices, what should asset 3 cost to prevent arbitrage? If it actually costs \$2.50, describe an arbitrage strategy.

Solution

Method 1: Solve state prices directly, then price claims

Let the state price vector be $q = (q_G, q_B)$. For any payoff $x = (x_G, x_B)$,

$$p(x) = q_G x_G + q_B x_B.$$

- Find** (q_G, q_B) .

From assets 1 and 2:

$$2q_G + q_B = 1.30, \quad q_G + 2q_B = 1.60.$$

Solve:

$$\begin{aligned} 2(q_G + 2q_B) - (2q_G + q_B) &= 3.20 - 1.30 \\ 3q_B &= 1.90 \end{aligned} \Rightarrow q_B = \frac{19}{30}.$$

Then

$$q_G = 1.60 - 2q_B = \frac{8}{5} - \frac{19}{15} = \frac{1}{3}.$$

Hence

$$(q_G, q_B) = \left(\frac{1}{3}, \frac{19}{30} \right).$$

- Price of contingent bundle** $(5, 4)$.

$$p(5, 4) = 5q_G + 4q_B = 5 \cdot \frac{1}{3} + 4 \cdot \frac{19}{30} = \frac{5}{3} + \frac{76}{30} = \frac{126}{30} = \frac{21}{5}.$$

Thus

$$p(5, 4) = \frac{21}{5} = 4.20.$$

(c) **Asset 3 no-arbitrage price and arbitrage if priced at \$2.50.**

Asset 3 payoff is $(3, 3)$. Its no-arbitrage price is

$$p_3^* = 3q_G + 3q_B = 3(q_G + q_B) = 3\left(\frac{1}{3} + \frac{19}{30}\right) = \frac{29}{10} = 2.90.$$

So

$$p_3^* = 2.90.$$

If actual price is $p_3 = 2.50 < 2.90$, asset 3 is underpriced.

Replicate $(3, 3)$ using assets 1 and 2:

$$\theta_1(2, 1) + \theta_2(1, 2) = (3, 3).$$

This gives

$$2\theta_1 + \theta_2 = 3, \quad \theta_1 + 2\theta_2 = 3 \Rightarrow \theta_1 = \theta_2 = 1.$$

Replicating cost:

$$1.30 + 1.60 = 2.90.$$

Arbitrage strategy:

- Buy 1 unit of asset 3 (pay 2.50).
- Short 1 unit each of asset 1 and asset 2 (receive $1.30 + 1.60 = 2.90$).

Net cashflow today:

$$+2.90 - 2.50 = \boxed{+0.40}.$$

State-wise future payoff:

$$(3, 3) - [(2, 1) + (1, 2)] = (3, 3) - (3, 3) = (0, 0).$$

Thus this is an arbitrage.

Method 2: Matrix form

Let

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}, \quad q = \begin{pmatrix} q_G \\ q_B \end{pmatrix}, \quad p = \begin{pmatrix} 1.30 \\ 1.60 \end{pmatrix}.$$

Then

$$Aq = p, \quad A^{-1} = \frac{1}{3} \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}.$$

Hence

$$q = A^{-1}p = \frac{1}{3} \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix} \begin{pmatrix} 1.30 \\ 1.60 \end{pmatrix} = \begin{pmatrix} 1/3 \\ 19/30 \end{pmatrix}.$$

So

$$q = \left(\frac{1}{3}, \frac{19}{30} \right).$$

(a)

$$(q_G, q_B) = \left(\frac{1}{3}, \frac{19}{30} \right).$$

(b) For $x = (5, 4)$,

$$p(x) = q^\top x = (1/3 \quad 19/30) \begin{pmatrix} 5 \\ 4 \end{pmatrix} = \frac{21}{5} = 4.20.$$

Thus

$$p(5, 4) = 4.20.$$

(c) For asset 3, $x_3 = (3, 3)$:

$$p_3^* = q^\top x_3 = 3(q_G + q_B) = \frac{29}{10} = 2.90.$$

If $p_3 = 2.50$, then $p_3 < p_3^*$, so long x_3 , short its replicating portfolio. Since

$$x_3 = (2, 1) + (1, 2),$$

arbitrage is:

Long asset 3, short asset 1, short asset 2

with initial profit

$$2.90 - 2.50 = \boxed{0.40},$$

and zero terminal payoff in both states.

Method 3: Risk-free discount factor + risk-neutral probabilities

Let

$$q_f := q_G + q_B = \frac{1}{3} + \frac{19}{30} = \frac{29}{30}.$$

Then q_f is the price of the risk-free payoff $(1, 1)$. Define risk-neutral probabilities:

$$\tilde{\pi}_G = \frac{q_G}{q_f}, \quad \tilde{\pi}_B = \frac{q_B}{q_f}, \quad \tilde{\pi}_G + \tilde{\pi}_B = 1.$$

Hence any payoff $x = (x_G, x_B)$ has price

$$p(x) = q_f (\tilde{\pi}_G x_G + \tilde{\pi}_B x_B).$$

(a) Using Method 1/2 values,

$$(q_G, q_B) = \left(\frac{1}{3}, \frac{19}{30} \right).$$

(Equivalently, $q_f = \frac{29}{30}$ and $\tilde{\pi}_G = \frac{10}{29}$, $\tilde{\pi}_B = \frac{19}{29}$.)

(b)

$$p(5, 4) = q_G \cdot 5 + q_B \cdot 4 = \frac{21}{5} = 4.20.$$

So

$$p(5, 4) = 4.20.$$

(c) Since asset 3 pays $(3, 3) = 3(1, 1)$,

$$p_3^* = 3q_f = 3 \cdot \frac{29}{30} = \frac{29}{10} = 2.90.$$

If $p_3 = 2.50$, it is underpriced. Buy asset 3 and short the replicating risk-equivalent payoff $(3, 3)$ formed by shorting one unit each of assets 1 and 2. Initial gain:

$$0.40,$$

future payoff = 0 in both states.

Marking scheme (suggested, 8 marks)

(a) **(3 marks)** State prices

- 1 mark: correct pricing equations.
- 1 mark: correct algebra/solution method.
- 1 mark: correct state prices $q_G = \frac{1}{3}, q_B = \frac{19}{30}$.

(b) **(2 marks)** Price of (5, 4)

- 1 mark: correct pricing formula $p = q_G x_G + q_B x_B$.
- 1 mark: correct value 4.20 .

(c) **(3 marks)** Asset 3 and arbitrage

- 1 mark: correct no-arbitrage price 2.90 .
- 1 mark: identify underpricing at 2.50.
- 1 mark: valid arbitrage strategy with state-wise cancellation and initial profit 0.40 .

Part II

Long Questions

Question 7: Log utility demand, endowments, and substitution effects

Consider a consumer with utility function:

$$U(x_1, x_2) = 2 \ln(x_1) + 3 \ln(x_2)$$

- (a) (4 marks) Show that the marginal utilities are $MU_1 = \frac{2}{x_1}$ and $MU_2 = \frac{3}{x_2}$. Find the marginal rate of substitution $MRS_{1,2}$ and write down the condition for utility maximization when facing prices (p_1, p_2) .
- (b) (6 marks) Derive the consumer's demand functions $x_1(p_1, p_2, m)$ and $x_2(p_1, p_2, m)$ where m is income.
- (c) (4 marks) The consumer has endowment $\omega = (10, 5)$ and faces prices $(p_1, p_2) = (3, 2)$. Calculate:
- The value of the endowment (income m)
 - The optimal consumption bundle
 - Whether the consumer is a net buyer or seller of each good
- (d) (3 marks) Verify that Walras' Law holds: $p \cdot x(p, p \cdot \omega) = p \cdot \omega$.
- (e) (3 marks) At the prices in part (c), suppose the price of good 1 increases to $p'_1 = 4$. Calculate the substitution effect Δx_1^s for good 1. Is good 1 a normal good?

Solution

Method 1: Economic interpretation (MU, MRS, budget shares, Slutsky)

(a) Marginal utilities and MRS.

$$MU_1 = \frac{\partial U}{\partial x_1} = \frac{2}{x_1}, \quad MU_2 = \frac{\partial U}{\partial x_2} = \frac{3}{x_2}.$$

Thus the marginal rate of substitution (good 1 for good 2) is

$$MRS_{1,2} = \frac{MU_1}{MU_2} = \frac{2/x_1}{3/x_2} = \frac{2x_2}{3x_1}.$$

At an interior optimum, utility maximization requires

$$\boxed{MRS_{1,2} = \frac{p_1}{p_2}} \iff \boxed{\frac{2x_2}{3x_1} = \frac{p_1}{p_2}},$$

together with the budget constraint

$$\boxed{p_1x_1 + p_2x_2 = m}.$$

(b) Marshallian demand functions.

For log/Cobb–Douglas utility $2 \ln x_1 + 3 \ln x_2$, optimal expenditure shares are constant:

$$\text{share on good 1} = \frac{2}{2+3} = \frac{2}{5}, \quad \text{share on good 2} = \frac{3}{2+3} = \frac{3}{5}.$$

Hence

$$p_1x_1 = \frac{2}{5}m, \quad p_2x_2 = \frac{3}{5}m,$$

so

$$\boxed{x_1(p_1, p_2, m) = \frac{2}{5} \frac{m}{p_1}, \quad x_2(p_1, p_2, m) = \frac{3}{5} \frac{m}{p_2}.$$

(c) Endowment income, optimal bundle, and net trade.

Given $\omega = (10, 5)$ and $p = (3, 2)$, income is the value of the endowment:

$$m = p \cdot \omega = 3(10) + 2(5) = 40.$$

Optimal consumption:

$$x_1^* = \frac{2}{5} \frac{40}{3} = \frac{16}{3}, \quad x_2^* = \frac{3}{5} \frac{40}{2} = 12.$$

Net trades:

$$x_1^* - \omega_1 = \frac{16}{3} - 10 = -\frac{14}{3} < 0 \implies \text{net seller of good 1,}$$

$$x_2^* - \omega_2 = 12 - 5 = 7 > 0 \implies \text{net buyer of good 2.}$$

Therefore

$$\boxed{m = 40, \quad x^* = \left(\frac{16}{3}, 12\right), \quad \text{sell good 1, buy good 2.}}$$

(d) **Walras' Law.**

$$p \cdot x(p, p \cdot \omega) = 3 \cdot \frac{16}{3} + 2 \cdot 12 = 16 + 24 = 40 = p \cdot \omega.$$

Hence

$$\boxed{p \cdot x(p, p \cdot \omega) = p \cdot \omega.}$$

(e) **Substitution effect for x_1 and normality.**

Initial bundle at $p = (3, 2)$, $m = 40$:

$$x = \left(\frac{16}{3}, 12 \right).$$

New price vector $p' = (4, 2)$. Slutsky compensated income:

$$m^s = p' \cdot x = 4 \cdot \frac{16}{3} + 2 \cdot 12 = \frac{64}{3} + 24 = \frac{136}{3}.$$

Compensated demand for good 1:

$$x_1^s = \frac{2}{5} \frac{m^s}{4} = \frac{2}{5} \cdot \frac{136/3}{4} = \frac{68}{15}.$$

Therefore

$$\Delta x_1^s = x_1^s - x_1 = \frac{68}{15} - \frac{16}{3} = \frac{68}{15} - \frac{80}{15} = -\frac{12}{15} = -\frac{4}{5}.$$

So

$$\boxed{\Delta x_1^s = -\frac{4}{5}.$$

Also,

$$\frac{\partial x_1}{\partial m} = \frac{2}{5p_1} > 0,$$

so good 1 is normal.

$$\boxed{\text{Good 1 is a normal good.}}$$

Method 2: Lagrangian derivation

(a)

$$MU_1 = \frac{2}{x_1}, \quad MU_2 = \frac{3}{x_2}, \quad MRS_{1,2} = \frac{MU_1}{MU_2} = \frac{2x_2}{3x_1}.$$

Interior optimum:

$$\frac{2x_2}{3x_1} = \frac{p_1}{p_2}, \quad p_1x_1 + p_2x_2 = m.$$

(b) Let

$$\mathcal{L}(x_1, x_2, \lambda) = 2 \ln x_1 + 3 \ln x_2 + \lambda(m - p_1x_1 - p_2x_2).$$

FOCs:

$$\frac{2}{x_1} = \lambda p_1, \quad \frac{3}{x_2} = \lambda p_2, \quad p_1x_1 + p_2x_2 = m.$$

Hence

$$x_1 = \frac{2}{\lambda p_1}, \quad x_2 = \frac{3}{\lambda p_2}.$$

Substitute into the budget:

$$\frac{2}{\lambda} + \frac{3}{\lambda} = m \Rightarrow \lambda = \frac{5}{m}.$$

Therefore

$$\boxed{x_1 = \frac{2m}{5p_1}, \quad x_2 = \frac{3m}{5p_2}.$$

(c) At $p = (3, 2)$, $\omega = (10, 5)$,

$$m = p \cdot \omega = 3(10) + 2(5) = 40.$$

Thus

$$x^*(p, m) = \left(\frac{2m}{5p_1}, \frac{3m}{5p_2} \right) = \left(\frac{2 \cdot 40}{5 \cdot 3}, \frac{3 \cdot 40}{5 \cdot 2} \right) = \left(\frac{16}{3}, 12 \right).$$

Net trade:

$$z = x - \omega = \left(\frac{16}{3} - 10, 12 - 5 \right) = \left(-\frac{14}{3}, 7 \right).$$

Hence good 1 is sold, good 2 is bought.

(d)

$$p \cdot x(p, p \cdot \omega) = p_1 \frac{2}{5} \frac{p \cdot \omega}{p_1} + p_2 \frac{3}{5} \frac{p \cdot \omega}{p_2} = \left(\frac{2}{5} + \frac{3}{5} \right) (p \cdot \omega) = p \cdot \omega.$$

So Walras' Law holds.

(e) Let $p = (3, 2)$, $p' = (4, 2)$, $m = 40$, and $x = (16/3, 12)$. Slutsky compensated income:

$$m^s = p' \cdot x = \frac{136}{3}.$$

Compensated demand:

$$x_1^s = \frac{2}{5} \frac{m^s}{4} = \frac{68}{15}.$$

Hence

$$\Delta x_1^s = x_1^s - x_1 = \frac{68}{15} - \frac{16}{3} = -\frac{4}{5}.$$

And

$$\frac{\partial x_1}{\partial m} = \frac{2}{5p_1} > 0$$

so good 1 is normal.

Marking scheme (suggested, 20 marks)(a) **(4 marks)** MU, MRS, utility-maximization condition

- 1 mark: correct $MU_1 = \frac{2}{x_1}$
- 1 mark: correct $MU_2 = \frac{3}{x_2}$
- 1 mark: correct $MRS_{1,2} = \frac{2x_2}{3x_1}$
- 1 mark: correct condition $MRS = \frac{p_1}{p_2}$ with budget constraint

(b) **(6 marks)** Demand derivation

- 2 marks: correct optimization setup / FOCs
- 2 marks: correct algebra to solve for λ or expenditure shares
- 2 marks: correct demands

$$x_1 = \frac{2}{5} \frac{m}{p_1}, \quad x_2 = \frac{3}{5} \frac{m}{p_2}$$

(c) **(4 marks)** Endowment and net trades

- 1 mark: correct income $m = 40$
- 1 mark: correct $x_1^* = \frac{16}{3}$
- 1 mark: correct $x_2^* = 12$
- 1 mark: correct buyer/seller classification

(d) **(3 marks)** Walras' Law

- 1 mark: correct substitution into $p \cdot x(p, p \cdot \omega)$
- 1 mark: correct calculation
- 1 mark: correct conclusion $= p \cdot \omega$

(e) **(3 marks)** Substitution effect and normality

- 1 mark: correct Slutsky compensated income $m^s = \frac{136}{3}$
- 1 mark: correct $\Delta x_1^s = \boxed{-\frac{4}{5}}$
- 1 mark: correct normality conclusion via $\partial x_1 / \partial m > 0$

Question 8: Intertemporal choice and Slutsky decomposition

Consider an intertemporal consumption problem with two periods (period 0 and period 1). A consumer has utility function:

$$U(x_0, x_1) = x_0^{0.4} x_1^{0.6}$$

The consumer can borrow and save at interest rate $r = 0.25$.

- (a) (3 marks) Write down the intertemporal budget constraint in present value form. Explain what each term represents.
- (b) (5 marks) Derive the consumer's optimal consumption demands $x_0^*(r, m)$ and $x_1^*(r, m)$ where m is the present value of lifetime income. Show that $x_1^*(r, m) = 1.5(1+r)x_0^*(r, m)$.
- (c) (4 marks) The consumer has income stream $(y_0, y_1) = (20, 50)$. At $r = 0.25$:
 - Calculate the present value of income m
 - Find the optimal consumption in each period
 - Determine whether the consumer is a net saver or borrower in period 0, and by how much
- (d) (4 marks) Verify that the budget constraint is satisfied at the optimal consumption bundle from part (c).
- (e) (4 marks) Now suppose the interest rate changes to $r' = 0.50$. Using Slutsky decomposition, calculate:
 - The Slutsky compensation m^s
 - The substitution effect Δx_0^s
 - The income effect Δx_0^i
 - Is period-0 consumption a normal good?

Solution

Method 1: Saving/borrowing choice s (one-variable approach)

Let s denote period-0 saving ($s < 0$ means borrowing). Then

$$x_0 = y_0 - s, \quad x_1 = y_1 + (1 + r)s.$$

(a) Substituting s into the PV relation gives the same budget:

$$x_0 + \frac{x_1}{1 + r} = y_0 + \frac{y_1}{1 + r} = m.$$

(b) The consumer solves

$$\max_s (y_0 - s)^{0.4} (y_1 + (1 + r)s)^{0.6}.$$

Taking logs:

$$\phi(s) = 0.4 \ln(y_0 - s) + 0.6 \ln(y_1 + (1 + r)s).$$

FOC:

$$-\frac{0.4}{y_0 - s} + \frac{0.6(1 + r)}{y_1 + (1 + r)s} = 0.$$

Using $x_0 = y_0 - s$, $x_1 = y_1 + (1 + r)s$, this becomes

$$-\frac{0.4}{x_0} + \frac{0.6(1 + r)}{x_1} = 0 \implies x_1 = 1.5(1 + r)x_0.$$

Together with $x_0 + \frac{x_1}{1 + r} = m$, this yields

$$\boxed{x_0^* = 0.4m, \quad x_1^* = 0.6(1 + r)m.}$$

(c) At $r = 0.25$:

$$m = 20 + \frac{50}{1.25} = 60, \quad (x_0^*, x_1^*) = (24, 45).$$

Thus

$$s = 20 - 24 = -4,$$

so the consumer is a net borrower of 4.

(d) Budget check:

$$24 + \frac{45}{1.25} = 60.$$

(e) At $r' = 0.50$, the Slutsky compensation and decomposition are identical to Methods 1–2:

$$\boxed{m^s = 54, \quad \Delta x_0^s = -\frac{12}{5}, \quad \Delta x_0^n = -\frac{4}{15}.$$

Also $\partial x_0^* / \partial m = 0.4 > 0$, so period-0 consumption is normal.

Method 2: Present-value Lagrangian (economic interpretation + Slutsky decomposition)**(a) Intertemporal budget constraint (present value form).**

Let m denote present value (PV) lifetime income. Then the PV budget constraint is

$$x_0 + \frac{x_1}{1+r} = m.$$

Interpretation:

- x_0 : period-0 consumption (already in present value)
- $\frac{x_1}{1+r}$: present value of period-1 consumption
- m : present value of lifetime resources (income/endowment stream)

If the income stream is (y_0, y_1) , then

$$m = y_0 + \frac{y_1}{1+r}.$$

(b) Optimal demands $x_0^*(r, m), x_1^*(r, m)$.

Solve

$$\max_{x_0, x_1 > 0} x_0^{0.4} x_1^{0.6} \quad \text{s.t.} \quad x_0 + \frac{x_1}{1+r} = m.$$

Equivalently maximize $0.4 \ln x_0 + 0.6 \ln x_1$. Lagrangian:

$$\mathcal{L} = 0.4 \ln x_0 + 0.6 \ln x_1 + \lambda \left(m - x_0 - \frac{x_1}{1+r} \right).$$

FOCs:

$$\frac{0.4}{x_0} = \lambda, \quad \frac{0.6}{x_1} = \frac{\lambda}{1+r}.$$

Substitute $\lambda = \frac{0.4}{x_0}$ into the second FOC:

$$\frac{0.6}{x_1} = \frac{0.4}{x_0(1+r)} \implies x_1 = \frac{0.6}{0.4}(1+r)x_0 = 1.5(1+r)x_0.$$

Hence

$$x_1^*(r, m) = 1.5(1+r)x_0^*(r, m).$$

Use the budget:

$$x_0 + \frac{1.5(1+r)x_0}{1+r} = x_0 + 1.5x_0 = 2.5x_0 = m$$

so

$$x_0^*(r, m) = \frac{m}{2.5} = 0.4m.$$

Then

$$x_1^*(r, m) = 1.5(1 + r)x_0^* = 1.5(1 + r)(0.4m) = 0.6(1 + r)m.$$

Therefore,

$$x_0^*(r, m) = 0.4m, \quad x_1^*(r, m) = 0.6(1 + r)m,$$

and indeed

$$x_1^*(r, m) = 1.5(1 + r)x_0^*(r, m).$$

- (c) **Numerical solution for** $(y_0, y_1) = (20, 50)$, $r = 0.25$.

Present value of income:

$$m = 20 + \frac{50}{1.25} = 20 + 40 = 60.$$

Optimal consumption:

$$x_0^* = 0.4m = 24, \quad x_1^* = 0.6(1.25)m = 0.75(60) = 45.$$

Saving in period 0:

$$s = y_0 - x_0^* = 20 - 24 = -4.$$

Since $s < 0$, the consumer is a *borrower* (borrows 4 units in period 0).

Thus

$$m = 60, \quad (x_0^*, x_1^*) = (24, 45), \quad \text{net borrower of 4 in period 0.}$$

- (d) **Verify the budget at the optimum.**

$$x_0^* + \frac{x_1^*}{1 + r} = 24 + \frac{45}{1.25} = 24 + 36 = 60 = m.$$

Hence the optimal bundle satisfies the budget exactly:

$$x_0^* + \frac{x_1^*}{1 + r} = m.$$

- (e) **Interest rate rises to $r' = 0.50$: Slutsky decomposition for x_0 .**

Old optimum (at $r = 0.25$) is $(x_0, x_1) = (24, 45)$.

Step 1: Slutsky compensation m^s . At the new interest rate $r' = 0.50$, the PV price of period-1 consumption is $1/(1 + r') = 1/1.5 = 2/3$. Slutsky-compensated income (cost of old bundle at new prices):

$$m^s = x_0 + \frac{x_1}{1 + r'} = 24 + \frac{45}{1.5} = 24 + 30 = 54.$$

So

$$m^s = 54.$$

Step 2: Compensated (Slutsky) demand at (r', m^s) . Since $x_0^*(r, m) = 0.4m$, at (r', m^s) :

$$x_0^s = 0.4m^s = 0.4(54) = 21.6 = \frac{108}{5}.$$

Substitution effect:

$$\Delta x_0^s = x_0^s - x_0 = \frac{108}{5} - 24 = \frac{108 - 120}{5} = -\frac{12}{5}.$$

Hence

$$\Delta x_0^s = -\frac{12}{5}.$$

Step 3: New Marshallian demand and income effect.

New present-value income:

$$m' = 20 + \frac{50}{1.5} = 20 + \frac{100}{3} = \frac{160}{3}.$$

New Marshallian period-0 consumption:

$$x'_0 = 0.4m' = \frac{2}{5} \cdot \frac{160}{3} = \frac{64}{3}.$$

Income effect:

$$\Delta x_0^n = x'_0 - x_0^s = \frac{64}{3} - \frac{108}{5} = \frac{320 - 324}{15} = -\frac{4}{15}.$$

Thus

$$\Delta x_0^n = -\frac{4}{15}.$$

Check total change:

$$\Delta x_0 = x'_0 - x_0 = \frac{64}{3} - 24 = -\frac{8}{3}$$

and

$$\Delta x_0^s + \Delta x_0^n = -\frac{12}{5} - \frac{4}{15} = -\frac{36 + 4}{15} = -\frac{40}{15} = -\frac{8}{3}.$$

Normality.

$$x_0^*(r, m) = 0.4m \quad \Rightarrow \quad \frac{\partial x_0^*}{\partial m} = 0.4 > 0.$$

So period-0 consumption is a normal good (with respect to present-value income m):

$$\boxed{\text{Period-0 consumption is normal.}}$$

Method 3: Pure Lagrangian derivation

(a)

$$x_0 + \frac{x_1}{1+r} = m, \quad m = y_0 + \frac{y_1}{1+r}.$$

(b)

$$\mathcal{L} = 0.4 \ln x_0 + 0.6 \ln x_1 + \lambda \left(m - x_0 - \frac{x_1}{1+r} \right).$$

FOCs:

$$\frac{0.4}{x_0} = \lambda, \quad \frac{0.6}{x_1} = \frac{\lambda}{1+r}, \quad x_0 + \frac{x_1}{1+r} = m.$$

Hence

$$\frac{0.6}{x_1} = \frac{0.4}{x_0(1+r)} \implies x_1 = 1.5(1+r)x_0.$$

Substitute into the constraint:

$$x_0 + \frac{1.5(1+r)x_0}{1+r} = 2.5x_0 = m \Rightarrow x_0^* = 0.4m.$$

Therefore

$$x_0^*(r, m) = 0.4m, \quad x_1^*(r, m) = 0.6(1+r)m,$$

and

$$x_1^*(r, m) = 1.5(1+r)x_0^*(r, m).$$

(c) With $(y_0, y_1) = (20, 50)$, $r = 0.25$:

$$m = 20 + \frac{50}{1.25} = 60,$$

$$x_0^* = 0.4(60) = 24, \quad x_1^* = 0.6(1.25)(60) = 45.$$

Saving:

$$s = y_0 - x_0^* = 20 - 24 = -4.$$

So the consumer borrows 4 in period 0.

(d)

$$x_0^* + \frac{x_1^*}{1+r} = 24 + \frac{45}{1.25} = 60 = m.$$

(e) Let $r' = 0.50$, old bundle $(24, 45)$.

$$m^s = 24 + \frac{45}{1.5} = 54, \quad x_0^s = 0.4(54) = \frac{108}{5}.$$

So

$$\Delta x_0^s = \frac{108}{5} - 24 = -\frac{12}{5}.$$

New PV income:

$$m' = 20 + \frac{50}{1.5} = \frac{160}{3}, \quad x'_0 = 0.4m' = \frac{64}{3}.$$

Hence

$$\Delta x_0^n = x'_0 - x_0^s = \frac{64}{3} - \frac{108}{5} = -\frac{4}{15}.$$

And

$$\frac{\partial x_0^*}{\partial m} = 0.4 > 0 \quad \Rightarrow \quad x_0 \text{ is normal.}$$

Marking scheme (suggested, 20 marks)(a) **(3 marks)** Intertemporal budget constraint (PV form)

- 1 mark: correct PV budget $x_0 + \frac{x_1}{1+r} = m$
- 1 mark: correct identification of m as PV lifetime income/resources
- 1 mark: correct explanation of terms (especially $\frac{x_1}{1+r}$)

(b) **(5 marks)** Optimal demands and ratio condition

- 2 marks: correct optimization setup / FOCs
- 1 mark: correct derivation of $x_1 = 1.5(1+r)x_0$
- 1 mark: correct $x_0^*(r, m) = 0.4m$
- 1 mark: correct $x_1^*(r, m) = 0.6(1+r)m$

(c) **(4 marks)** Numerical solution at $r = 0.25$

- 1 mark: correct $m = 60$
- 1 mark: correct $x_0^* = 24$
- 1 mark: correct $x_1^* = 45$
- 1 mark: correct borrower/saver classification and amount (borrow 4)

(d) **(4 marks)** Budget verification

- 2 marks: correct substitution into budget
- 2 marks: correct equality $= m$ and conclusion

(e) **(4 marks)** Slutsky decomposition for $r' : 0.25 \rightarrow 0.50$

- 1 mark: correct Slutsky compensation $m^s = 54$
- 1 mark: correct substitution effect $\Delta x_0^s = -\frac{12}{5}$
- 1 mark: correct income effect $\Delta x_0^i = -\frac{4}{15}$
- 1 mark: correct normality conclusion ($\partial x_0 / \partial m > 0$)

Question 9: State prices, optimal contingent claims, and replication

Consider an economy with two states of nature (state 1 and state 2) with probabilities $\pi_1 = \frac{1}{3}$ and $\pi_2 = \frac{2}{3}$. There are two assets:

- Risk-free asset: pays \$1 in both states and costs p_f
- Risky asset: pays \$6 in state 1 and \$0 in state 2, and costs $p_r = \$1.50$

A consumer has expected utility preferences with $U(x_1, x_2) = \frac{1}{3}u(x_1) + \frac{2}{3}u(x_2)$ where $u(x) = \ln(x)$.

- (4 marks) The consumer is endowed with $(0, 9)$ across the two states. If she can trade the risky asset and contingent consumption can be purchased at state prices $(q_1, q_2) = (0.25, 0.50)$:
 - What is the value of her endowment?
 - What is the implied price p_r^* of the risky asset using state prices?
 - Is the actual price $p_r = \$1.50$ consistent with no arbitrage?
- (6 marks) Using the state prices from part (a), find the consumer's optimal contingent consumption bundle (x_1^*, x_2^*) . [Hint: At optimum, $\frac{\pi_1 u'(x_1)}{\pi_2 u'(x_2)} = \frac{q_1}{q_2}$]
- (4 marks) The consumer wants to achieve the optimal bundle from part (b) using only the two assets (risk-free and risky). Set up and solve the system of equations to find the portfolio (θ_f, θ_r) where θ_f is units of risk-free asset and θ_r is units of risky asset.
- (3 marks) Verify that the cost of the portfolio from part (c) equals the cost of the contingent consumption bundle from part (b) using state prices.
- (3 marks) Suppose a new asset becomes available that pays \$3 in state 1 and \$2 in state 2. What should this asset cost to prevent arbitrage opportunities? If it costs \$1.80, is there an arbitrage opportunity? Describe the strategy if yes.

Solution

Method 1: Arrow–Debreu pricing + expected utility + replication

Let the state prices be

$$(q_1, q_2) = (0.25, 0.50) = \left(\frac{1}{4}, \frac{1}{2}\right).$$

For any contingent claim $x = (x_1, x_2)$, its Arrow–Debreu price is

$$p(x) = q_1 x_1 + q_2 x_2.$$

(a) Endowment value and risky asset pricing.

Endowment $\omega = (0, 9)$ has value

$$m = q \cdot \omega = q_1 \omega_1 + q_2 \omega_2 = \frac{1}{4}(0) + \frac{1}{2}(9) = \frac{9}{2} = 4.5.$$

So

$$\boxed{m = 4.5.}$$

Risky asset payoff is $(6, 0)$, so the implied no-arbitrage price is

$$p_r^* = 6q_1 + 0q_2 = 6 \cdot \frac{1}{4} = \frac{3}{2} = 1.50.$$

Thus

$$\boxed{p_r^* = 1.50.}$$

Since $p_r = 1.50 = p_r^*$, the quoted risky-asset price is consistent with no arbitrage:

$$\boxed{\text{Yes, } p_r = 1.50 \text{ is no-arbitrage consistent.}}$$

(b) Optimal contingent consumption (x_1^*, x_2^*) .

The consumer solves

$$\max_{x_1, x_2 > 0} \frac{1}{3} \ln x_1 + \frac{2}{3} \ln x_2 \quad \text{s.t.} \quad q_1 x_1 + q_2 x_2 = m.$$

Here $m = 4.5$, $q_1 = \frac{1}{4}$, $q_2 = \frac{1}{2}$.

Lagrangian:

$$\mathcal{L} = \frac{1}{3} \ln x_1 + \frac{2}{3} \ln x_2 + \lambda(m - q_1 x_1 - q_2 x_2).$$

FOCs:

$$\frac{1}{3x_1} = \lambda q_1, \quad \frac{2}{3x_2} = \lambda q_2.$$

Divide:

$$\frac{\frac{1}{3x_1}}{\frac{2}{3x_2}} = \frac{q_1}{q_2} \implies \frac{x_2}{2x_1} = \frac{q_1}{q_2} = \frac{1/4}{1/2} = \frac{1}{2} \implies x_2 = x_1.$$

Use the budget:

$$\frac{1}{4}x_1 + \frac{1}{2}x_2 = 4.5.$$

Since $x_2 = x_1$,

$$\left(\frac{1}{4} + \frac{1}{2}\right)x_1 = \frac{3}{4}x_1 = 4.5 \implies x_1 = 6.$$

Hence $x_2 = 6$. Therefore

$$(x_1^*, x_2^*) = (6, 6).$$

(c) **Replicate** $(x_1^*, x_2^*) = (6, 6)$ **using the two assets.**

Payoff vectors:

$$a_f = (1, 1), \quad a_r = (6, 0).$$

Find (θ_f, θ_r) such that

$$\theta_f(1, 1) + \theta_r(6, 0) = (6, 6).$$

Equivalently,

$$\begin{cases} \theta_f + 6\theta_r = 6, \\ \theta_f = 6. \end{cases}$$

From the second equation, $\theta_f = 6$. Substituting into the first:

$$6 + 6\theta_r = 6 \implies \theta_r = 0.$$

Thus

$$(\theta_f, \theta_r) = (6, 0).$$

(d) **Verify portfolio cost = contingent-claim cost.**

First compute risk-free price from state prices:

$$p_f = q_1 + q_2 = \frac{1}{4} + \frac{1}{2} = \frac{3}{4} = 0.75.$$

Portfolio cost:

$$\theta_f p_f + \theta_r p_r = 6(0.75) + 0(1.50) = 4.5.$$

Contingent-claim cost (state-price cost):

$$q_1 x_1^* + q_2 x_2^* = \frac{1}{4}(6) + \frac{1}{2}(6) = 1.5 + 3 = 4.5.$$

Hence

$$\theta_f p_f + \theta_r p_r = q_1 x_1^* + q_2 x_2^* = 4.5.$$

(e) **Pricing a new asset with payoff (3, 2), and arbitrage if priced at 1.80.**

No-arbitrage price from state prices:

$$p_{\text{new}}^* = 3q_1 + 2q_2 = 3\left(\frac{1}{4}\right) + 2\left(\frac{1}{2}\right) = \frac{3}{4} + 1 = \frac{7}{4} = 1.75.$$

So

$$p_{\text{new}}^* = 1.75.$$

If the market price is $1.80 > 1.75$, the new asset is overpriced, so there is an arbitrage.

Replicate (3, 2) using old assets:

$$\theta_f(1, 1) + \theta_r(6, 0) = (3, 2)$$

i.e.

$$\begin{cases} \theta_f + 6\theta_r = 3, \\ \theta_f = 2. \end{cases}$$

Thus $\theta_f = 2$ and

$$2 + 6\theta_r = 3 \Rightarrow \theta_r = \frac{1}{6}.$$

Replicating portfolio cost:

$$2p_f + \frac{1}{6}p_r = 2(0.75) + \frac{1}{6}(1.50) = 1.50 + 0.25 = 1.75.$$

Arbitrage strategy (since market price 1.80 is too high):

- Short 1 unit of the new asset (receive 1.80 today),
- Buy the replicating portfolio $(\theta_f, \theta_r) = (2, \frac{1}{6})$ (pay 1.75 today).

Net cashflow today:

$$1.80 - 1.75 = \boxed{0.05}.$$

Future state payoffs cancel exactly, so this is arbitrage.

Yes, arbitrage exists if price = 1.80: short new asset, buy replica, profit 0.05 today.

Method 2: State-contingent Cobb–Douglas shares

(a) Endowment value:

$$m = q_1\omega_1 + q_2\omega_2 = 0.25(0) + 0.50(9) = 4.5.$$

Risky asset implied price:

$$p_r^* = 6q_1 = 1.5.$$

Hence $p_r = 1.50$ is no-arbitrage consistent.

(b) Since the objective is

$$\frac{1}{3} \ln x_1 + \frac{2}{3} \ln x_2 = \ln(x_1^{1/3} x_2^{2/3}),$$

this is Cobb–Douglas in state-contingent consumption with expenditure shares $1/3$ and $2/3$. Thus

$$q_1 x_1 = \frac{1}{3} m, \quad q_2 x_2 = \frac{2}{3} m.$$

With $m = 4.5$, $q_1 = 0.25$, $q_2 = 0.50$:

$$x_1 = \frac{(1/3)4.5}{0.25} = 6, \quad x_2 = \frac{(2/3)4.5}{0.50} = 6.$$

Hence

$$(x_1^*, x_2^*) = (6, 6).$$

(c) Replication of $(6, 6)$:

$$\theta_f(1, 1) + \theta_r(6, 0) = (6, 6) \Rightarrow (\theta_f, \theta_r) = (6, 0).$$

(d) Since $p_f = q_1 + q_2 = 0.75$,

$$\theta_f p_f + \theta_r p_r = 6(0.75) = 4.5$$

and

$$q_1 x_1^* + q_2 x_2^* = 0.25(6) + 0.50(6) = 4.5.$$

So costs coincide.

(e) New asset payoff $(3, 2)$ must cost

$$p^* = 3q_1 + 2q_2 = 1.75.$$

If priced at 1.80, it is overpriced; short it and buy its replicating portfolio $(2, \frac{1}{6})$, earning 0.05 today with zero future net payoff.

Method 3: Lagrangian

(a)

$$m = q \cdot \omega = \frac{1}{4}(0) + \frac{1}{2}(9) = \frac{9}{2}.$$

Risky asset no-arbitrage price:

$$p_r^* = q \cdot (6, 0) = 6 \cdot \frac{1}{4} = \frac{3}{2}.$$

Since $p_r = \frac{3}{2}$, no arbitrage violation.

(b)

$$\max_{x_1, x_2 > 0} \frac{1}{3} \ln x_1 + \frac{2}{3} \ln x_2 \quad \text{s.t.} \quad \frac{1}{4}x_1 + \frac{1}{2}x_2 = \frac{9}{2}.$$

$$\mathcal{L} = \frac{1}{3} \ln x_1 + \frac{2}{3} \ln x_2 + \lambda \left(\frac{9}{2} - \frac{1}{4}x_1 - \frac{1}{2}x_2 \right).$$

FOCs:

$$\frac{1}{3x_1} = \frac{\lambda}{4}, \quad \frac{2}{3x_2} = \frac{\lambda}{2}.$$

Eliminate λ :

$$\frac{1}{3x_1} \bigg/ \frac{2}{3x_2} = \frac{1/4}{1/2} \Rightarrow \frac{x_2}{2x_1} = \frac{1}{2} \Rightarrow x_2 = x_1.$$

Constraint:

$$\frac{3}{4}x_1 = \frac{9}{2} \Rightarrow x_1 = 6, \quad x_2 = 6.$$

Thus

$$(x_1^*, x_2^*) = (6, 6).$$

(c) Replication system:

$$\begin{pmatrix} 1 & 6 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} \theta_f \\ \theta_r \end{pmatrix} = \begin{pmatrix} 6 \\ 6 \end{pmatrix}.$$

From row 2, $\theta_f = 6$; then row 1 gives $\theta_r = 0$. Hence

$$(\theta_f, \theta_r) = (6, 0).$$

(d)

$$p_f = q_1 + q_2 = \frac{3}{4}.$$

$$\theta_f p_f + \theta_r p_r = 6 \cdot \frac{3}{4} + 0 \cdot \frac{3}{2} = \frac{9}{2} = q \cdot (6, 6).$$

(e) New asset payoff $a = (3, 2)$:

$$p^* = q \cdot a = 3 \cdot \frac{1}{4} + 2 \cdot \frac{1}{2} = \frac{7}{4}.$$

If market price is $1.80 = \frac{9}{5} > \frac{7}{4}$, asset is overpriced. Short new asset; buy replica solving

$$\theta_f(1, 1) + \theta_r(6, 0) = (3, 2) \Rightarrow (\theta_f, \theta_r) = \left(2, \frac{1}{6}\right).$$

Initial arbitrage profit:

$$\frac{9}{5} - \frac{7}{4} = \frac{36 - 35}{20} = \boxed{\frac{1}{20} = 0.05}.$$

Marking scheme (suggested, 20 marks)

(a) **(4 marks)** Endowment value and risky-asset pricing

- 1 mark: correct endowment valuation $m = q \cdot \omega$
- 1 mark: correct $m = 4.5$
- 1 mark: correct implied risky asset price $p_r^* = 1.50$
- 1 mark: correct no-arbitrage conclusion (actual price consistent)

(b) **(6 marks)** Optimal contingent consumption

- 2 marks: correct setup (EU maximization with state-price budget)
- 2 marks: correct FOCs / ratio condition and implication $x_2 = x_1$
- 2 marks: correct solution $\boxed{(x_1^*, x_2^*) = (6, 6)}$

(c) **(4 marks)** Replicating portfolio

- 2 marks: correct system of equations
- 2 marks: correct solution $\boxed{(\theta_f, \theta_r) = (6, 0)}$

(d) **(3 marks)** Cost verification

- 1 mark: correct $p_f = q_1 + q_2 = 0.75$
- 1 mark: correct portfolio cost
- 1 mark: correct state-price cost and equality

(e) **(3 marks)** New asset pricing and arbitrage

- 1 mark: correct no-arbitrage price $\boxed{1.75}$
- 1 mark: identify overpricing at 1.80
- 1 mark: correct arbitrage strategy and profit $\boxed{0.05}$